Relational Algebra

txt2tex example

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$[Title, Author, MemberName, Genre]$

Book

$bookId : \mathbb{N}$
$title : Title$
$author : Author$
$genre : Genre$
$pages : \mathbb{N}$
$year : \mathbb{N}$

Member

$memberId : \mathbb{N}$
$name : MemberName$

Loan

$loanId : \mathbb{N}$
$bookId : \mathbb{N}$
$memberId : \mathbb{N}$
$dueYear : \mathbb{N}$

Restriction (sigma)

$\text{Restrict}_{pages \geq 500}(Book)$
$\text{Restrict}_{year \geq 1990}(\text{Restrict}_{pages \geq 500}(Book))$

Projection (pi)

$\text{Project}_{title, author}(Book)$
$\text{Project}_{title, author}(\text{Restrict}_{pages \geq 500}(Book))$

Renaming (R[new/old])

$\text{Project}_{id, title, author, genre, pages, year}(Book[id / bookId])$
$\text{Project}_{id, writer, genre, pages, year}(Book[id / bookId, writer / author])$

Natural Join and Theta-Join (join)

$\text{Join}(Loan, Book)$
$\text{Project}_{id, memberId, dueYear, title, author, genre, pages, year}(\text{Join}(Loan[id / bookId], Book[id / bookId]))$
$\text{Join}_{Member.memberId = Loan.memberId}(Member, Loan)$

Division (div)

$Loan \ \mathsf{div} \ Book$

Combined Query

$\mathrm{Project}_{title,author}(\mathrm{Restrict}_{pages \geq 500}(Book))$